

NOTE · Integer Mellin Moments of the Dome

1, 7, 124, 4318, ... – Integer Mellin Moments of the Dome $\tanh(x)/\cosh(2x)$

Mersenne and Bernoulli Primes on the η -Branch of a Bad-Prime Modification of ζ

Title finalized 2026-06-10 – the sequence values lead the title deliberately: with no OEIS entry by decision (Acknowledgments), the title and abstract are the sequence's catalog entry.

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Reprinted 16 September 2026 (notice below)

► READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.

How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>

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UPDATE NOTICE – 16 September 2026. This reprint adds the three lines above this notice (a pointer to One coordinate up, where the corpus is now read); nothing else changes. The paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "Integer Mellin Moments of the Dome". Re-computed 2026-09-11 in exact arithmetic and at 40 digits: $a(1..10)$ from the Bernoulli form, the A002105 and A000182 forms against it, $M_g(s)$ by quadrature against the closed form, $a(n) = M(2n)/M(2)^n$ for $n = 1..5$, $c_n = (2n-1) \cdot |E_{2n-2}|$ for $n = 1..6$, and the positions of 691, 3617 and 43867 in $a(6)$, $a(8)$, $a(9)$: all stand. Prior version remains on OSF in the project's archive/ folder, named by its date. Started: 2026-06-05 Project notebook: Working_Notes_Dome_Moment_Sequence.md (running trace log; §1 stress-test traces, §2 verification log, §3 open questions, §3.A results ledger – Theorems T1–T9, T13, T15; 010, R11–R12, D14 – §4 decisions log) Cross-framework notebook: Cross_Framework_Connections_Notes.md (§A.1 pattern-search; §A.2 structural completion; §A.3 L-function classification)

Framework hub: gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html

Companion paper: Paper 11, The η Corridor in 3D Wilson-Fisher Critical Phenomena, DOI 10.17605/OSF.IO/PD73B.

Abstract

Let $g(x) = \tanh(x) / \cosh(2x)$ be the dome function – the bounded, even-Mellin-supported weight introduced in Paper 11 of the Gap Geometry framework. Its even Mellin moments produce, after ratio normalization, the integer sequence

$$a(n) = 1, 7, 124, 4318, 253456, 22631632, 2862787264, 487346998768, \dots$$

This paper documents three structural facts about $a(n)$. The integrality of $a(n)$ is proved for all $n \geq 1$ (Theorem T15: von Staudt–Clausen, the lifting-the-exponent lemma, and the p -integrality of $B_{2n}/2n$ when $(p-1) \nmid 2n$ – observed by Adams in 1878, proved later – close all prime cases); integrality is additionally verified at exact arithmetic for $n = 1..60$, and the closed forms are cross-verified against each other and against quadrature for $n = 1..20$. First, the sequence admits a clean two-factor closed form $a(n) = A002105(n) \cdot (2^{2n-1} - 1) -$ reduced tangent numbers multiplied by an odd-indexed Mersenne companion – equivalent to the Bernoulli form $(2^{2n-1} - 1)(2^{2n} - 1) \cdot |B_{2n}| \cdot 2^n / n$. Second, the underlying Mellin transform of $g(x)$ has the closed-form representation $M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot \zeta(s) \cdot (1 - 2^{(-s)})(1 - 2^{(1-s)})$, which is structurally the partial-zeta Euler product over odd primes with a specifically engineered local factor at the bad prime $p = 2$ – not a primitive Dirichlet L-function, but a bad-prime modification of ζ . Third, this dome occupies a distinguished position in the small-integer family $g_{\{a,b,c\}}(x) = \tanh^a(x) / \cosh^b(c \cdot x)$: within the full integer-coordinate cube $\{1 \leq a, b \leq 3, 1 \leq c \leq 3\}$ (all 27 triples classified), the triple (1, 1, 2) is the unique member whose normalized even-moment ratios are integers; its nearest

sibling $(1, 1, 1) = \text{sech}(x) \cdot \tanh(x)$ routes through Dirichlet β at odd arguments and produces the closed-form $c_n = (2n-1) \cdot |E_{2n-2}|$. Uniqueness in the whole family $g_{\{a,b,c\}}$ (for all positive integer triples) is a strictly larger statement and is not claimed here – T8 establishes uniqueness in the full $3 \times 3 \times 3$ cube; the broader classification over all positive integer triples is logged as an open structural target in §9.

The Mersenne and Bernoulli prime families co-appear in $a(n)$ as the visible arithmetic shadow of a single structural fact: the dome's Mellin spectrum is ζ with a surgically modified local Euler factor at $p = 2$. The integer character of the sequence is a downstream consequence of this η -branch routing; the central object is the bifurcation between the η image and the β image in the Mellin transform of the dome family. The classical Kummer-irregular primes ($691 \in B_{12}$, $3617 \in B_{16}$, $43867 \in B_{18}$, ...) appear in $a(n)$ at predictable positions by construction.

The outline below presents the closed forms, the structural reading, the bifurcation classification, the arithmetic-shadow factorization patterns, framework context, and a section of open questions that frame the result as a concrete entry point into the classification of Mellin-weight families by L-function image branch.

§1 – The sequence

THE DOME MOMENT SEQUENCE

1 · 7 · 124 · 4318 · 253456 · 22631632 · 2862787264 · 487346998768

(continuing)

107450235817216 · 29786961541311232 · . . .

– the framework's signature line, in the form used on the Gap Geometry entry page. The dots are deliberate: the sequence factors multiplicatively in two distinguishable ways (Mersenne-companion $\times$ A002105, or Mersenne-pair $\times$ Bernoulli $\times$ binary-normalization), and the visual form invites the reader to notice that before any analysis begins. The indexed presentation, extended to $n = 10$:

n :	a(n)
1 :	1
2 :	7
3 :	124
4 :	4318
5 :	253456
6 :	22631632
7 :	2862787264
8 :	487346998768
9 :	107450235817216
10 :	29786961541311232

These integers are the values of the normalized even-moment ratio

$$a(n) = M(2n) / M(2)^n$$

where $M(2k) = \int_0^\infty x^{(2k-1)} \cdot g(x) \, dx$ is the even Mellin moment of order $2k$ of the dome function $g(x) = \tanh(x) / \cosh(2x)$, evaluated at the integer rescaling $s = 2k$ and with the natural $\pi^{(2n)}$ factor cleared by the $M(2)^n$ denominator.

The sequence is fast-growing: $a(n+1)/a(n) \approx (32/\pi^2) \cdot n^2 \approx 3.24 \cdot n^2$ for moderate n (Theorem T6 below), so $a(n) \sim \exp(2n \log n)$ in leading order.

Each integer factorizes into a Mersenne-companion piece and an A002105 piece, with the irregular Bernoulli primes appearing inside the second factor at positions dictated by classical number theory. The point of this document is to make those structural pieces explicit.

§2 – The headline factorization

A note on labels: theorem numbers are ledger IDs from the working notebook's results ledger (reproduced in §8.2), not document order – T2 appears here, T13 in §4, T8 in §5.

Theorem T2. For all $n \geq 1$,

$$a(n) = A002105(n) \cdot (2^{(2n-1)} - 1)$$

where A002105 is the reduced tangent number sequence catalogued in OEIS:

A002105 : 1, 1, 4, 34, 496, 11056, 349504, 14873104, 819786496, 56814228736, ...

and $(2^{(2n-1)} - 1)$ is the odd-indexed Mersenne-like companion:

n	:	1	2	3	4	5	6	7	8	9	10
$2^{(2n-1)}-1$	:	1	7	31	127	511	2047	8191	32767	131071	524287

Verified at exact arithmetic for $n = 1..10$ in Working_Notes_Dome_Moment_Sequence.md §2.1. Independently re-derived from Theorem T1 (Bernoulli form, below) via Euler's formula $B_{\{2n\}} = 2^{(-1)^{(n+1)}} (2n)! \cdot \zeta(2n) / (2\pi)^{(2n)}$ and the standard tangent-number identity $A000182(n) = 2^{(2n-1)}(2^{(2n)}-1) \cdot |B_{\{2n\}}| / n$, with reduced tangent numbers $A002105(n) = A000182(n) / 2^{(n-1)} = 2^n \cdot (2^{(2n)}-1) \cdot |B_{\{2n\}}| / n$. (Label corrected 2026-06-12 – the formula was previously misattributed to A002105; caught by Sonnet in the pre-deposit round, confirmed at exact arithmetic.)

Equivalent closed form (Bernoulli form, T1).

$$a(n) = (2^{(2n-1)} - 1) \cdot (2^{(2n)} - 1) \cdot |B_{\{2n\}}| \cdot 2^n / n$$

This is the form first derived from the dome's Mellin transform (Paper 11 Theorem 9.7). It makes the Bernoulli number content explicit and is the bridge to the L-function reading of §3–§4.

Equivalent closed form (tangent form).

$$a(n) = A000182(n) \cdot (2^{(2n-1)} - 1) / 2^{(n-1)}$$

where A000182 is the tangent number sequence 1, 2, 16, 272, 7936, ... This form is the most direct combinatorial expression – A000182(n) counts alternating permutations on $(2n-1)$ letters – and $a(n)$ is therefore that combinatorial count, tilted by the Mersenne companion and divided by a binary normalization.

Verification status. All three forms produce identical integers for $n = 1..10$ at exact arithmetic; the three are algebraically equivalent and each makes a different structural content visible. T2 (the A002105 $\times$ Mersenne form) is the headline; T1 (the Bernoulli form) is the bridge to the L-function structure; the tangent form is the combinatorial alternate.

§3 – The source: the dome function and its Mellin transform

§3.1 The function

Define

$$g(x) = \tanh(x) / \cosh(2x) \text{ for } x \geq 0,$$

extended by symmetry as an even function on the real line. Then g is positive, peaks once on $(0, \infty)$, and decays as $g(x) \sim 2 e^{(-2x)} - 4 e^{(-4x)} + 2 e^{(-6x)} + O(e^{(-10x)})$ as $x \rightarrow \infty$ – the exponential expansion of §3.2 below; the $e^{(-8x)}$ coefficient vanishes identically. Alternative definition (doubling increment). The dome admits the equivalent one-line form

$$g(x) = \tanh(2x) - \tanh(x)$$

(elementary: $\tanh 2x - \tanh x = \tanh x \cdot (1 - \tanh^2 x) / (1 + \tanh^2 x) = \tanh x \cdot \operatorname{sech} 2x$). Two consequences follow at once. The dome integral telescopes – $\int_0^X [\tanh 2x - \tanh x] dx = \frac{1}{2} \ln(\cosh 2X / \cosh^2 X) \rightarrow \ln(2)/2$ as $X \rightarrow \infty$ – recovering Paper 11 Theorem 9.5 with no further machinery. And iterating the identity along the doubling ladder $x \rightarrow 2x \rightarrow 4x \rightarrow \dots$ telescopes the sum:

$$\sum_{k=0}^{\infty} g(2^k x) = 1 - \tanh(x) \quad (x > 0):$$

the dome, evaluated rung by rung up the doubling ladder, sums exactly to the saturation deficit. (Surfaced in the 2026-06-10 audit round from the (1, 2, 2) mixing law of §9.2; verified to 50 digits; working notebook §2.17.)

Within the Gap Geometry framework, g is the central weight function of Paper 11; this paper takes it as input and studies the arithmetic of its even Mellin moments only.

§3.2 The Mellin transform – closed form (Paper 11 Theorem 9.7)

The Mellin transform of g is

$$M_g(s) = \int_0^\infty x^{(s-1)} \cdot g(x) dx = \Gamma(s) \cdot \eta(s) \cdot (2^s - 1) / 2^{(2s-1)}$$

valid for s with $\operatorname{Re}(s) > 0$, where Γ is the gamma function – equal to $(s - 1)!$ at the positive integer arguments $s = 2n$ used throughout this paper – and η is the Dirichlet eta function

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{(n+1)} n^{(-s)} = (1 - 2^{(1-s)}) \cdot \zeta(s).$$

This closed form follows by term-by-term Mellin integration of the exponential expansion

$$g(x) = 2 \cdot \sum_{m=1}^{\infty} (-1)^{(m+1)} \cdot (e^{(-2mx)} - e^{(-4mx)})$$

(numerically verified to 40 digits; working notebook §2.13). Since the Mellin transform of $e^{(-ax)}$ is $\Gamma(s) \cdot a^{(-s)}$, termwise integration gives $M_g(s) = \Gamma(s) \cdot 2 \cdot \eta(s) \cdot (2^{(-s)} - 4^{(-s)})$, and $2 \cdot (2^{(-s)} - 4^{(-s)}) = (2^s - 1) / 2^{(2s-1)}$ recovers the closed form – the Dirichlet eta core appears directly from the alternating signs of the expansion. The full derivation is the Paper 11 calculation; this paper takes $M_g(s)$ as established and explores its consequences for the even-integer values $s = 2n$.

Notation note for readers arriving from Paper 11. The Greek letter η carries two completely unrelated meanings inside the Gap Geometry framework, and both appear in companion papers:

- In Paper 11, The η Corridor in 3D Wilson-Fisher Critical Phenomena, η is the anomalous dimension – a critical-phenomena exponent measuring deviation from free-field scaling in second-order phase transitions, characterized empirically as bounded by the framework constant $2G$ across ten universality classes.
- In this paper, η is the Dirichlet eta function $\eta(s) = \sum_{n \geq 1} (-1)^{(n+1)} / n^s = (1 - 2^{(1-s)}) \cdot \zeta(s)$ – an analytic L-function on the complex plane, the alternating-sign companion to the Riemann zeta function.

The two objects share a Greek letter and a framework, and nothing else. The “ η -branch” language used throughout this paper (§4, §5, §8, §9) refers to the Dirichlet eta function exclusively; readers should not import critical-phenomena meaning into it.

§3.3 The normalized even-moment ratio

The Mellin transform at integer $s = 2n$ produces, after clearing the $(2n-1)!$ and the $(2\pi)^{(2n)}$ hidden in $B_{\{2n\}}$, an explicit value involving $\pi^{(2n)}$. The ratio

$$a(n) = M(2n) / M(2)^n$$

clears this $\pi^{(2n)}$ factor exactly and isolates the integer arithmetic content. The relation to T1's Bernoulli form follows from substituting $B_{\{2n\}} = 2 (-1)^{(n+1)} (2n)! \cdot \zeta(2n) / (2\pi)^{(2n)}$ into the closed form for $M_g(2n)$ and simplifying – the π powers cancel ratio-wise, the $(2n-1)!$ folds into the $(2n)! / n$ in the Bernoulli denominator, and the $2^n / n$ factor emerges from the binary structure of $(2^s - 1) / 2^{(2s-1)}$ at $s = 2n$.

The sequence $a(n)$ is therefore the integer arithmetic shadow of the dome's analytic Mellin transform, viewed under the natural dimensionless normalization for an even-moment generating function.

§4 – Structural reading: ζ with engineered local factor at $p = 2$

This section presents the central structural fact about $M_g(s)$ – the fact that organizes everything in the paper, including the integer character of $a(n)$. The result is elementary algebraic manipulation of the Paper 11 closed form together with a Dirichlet-series computation; we present it here as a theorem because the consequences for the document's framing are substantial.

§4.1 Theorem T13 (η -to- ζ rewrite)

Theorem T13. The Mellin transform of the dome admits the equivalent expression

$$M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot \zeta(s) \cdot (1 - 2^{(-s)})(1 - 2^{(1-s)})$$

equivalently, viewed via the Euler product of ζ :

$$M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot (1 - 2^{(1-s)}) \cdot \prod_{\{p > 2\}} (1 - p^{(-s)})^{(-1)}.$$

Proof. Substitute the standard identity $\eta(s) = (1 - 2^{(1-s)}) \cdot \zeta(s)$ into the Paper 11 closed form. The factor $(2^s - 1) / 2^{(2s-1)}$ simplifies as

$$(2^s - 1) / 2^{(2s-1)} = 2 \cdot 2^{(-s)} \cdot (1 - 2^{(-s)}) = 2^{(1-s)} \cdot (1 - 2^{(-s)}).$$

Combining gives

$$M_g(s) = \Gamma(s) \cdot (1 - 2^{(1-s)}) \zeta(s) \cdot 2^{(1-s)} \cdot (1 - 2^{(-s)}) = \Gamma(s) \cdot 2^{(1-s)} \cdot \zeta(s) \cdot (1 - 2^{(-s)})(1 - 2^{(1-s)}).$$

Since ζ has Euler factor $(1 - p^{(-s)})^{(-1)}$ at every prime p , the factor $(1 - 2^{(-s)})$ in the product cancels ζ 's local Euler factor at $p = 2$ exactly, leaving

$$M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot (1 - 2^{(1-s)}) \cdot \prod_{\{p > 2\}} (1 - p^{(-s)})^{(-1)}.$$

■

The factor $(1 - 2^{(1-s)})$ is precisely the η -conversion factor at $p = 2$ – it is what η contributes at the prime 2, and what survives the cancellation of ζ 's natural local factor. So in the rewritten form the $p = 2$ local factor of M_g (modulo the $\Gamma(s) \cdot 2^{(1-s)}$ prefactor) is $(1 - 2^{(1-s)})$, while every other local factor matches ζ .

§4.2 What this says about M_g

The Mellin transform of the dome is, in Euler-product language, ζ with the local Euler factor at $p = 2$ specifically modified. It is not the L-function of a primitive Dirichlet character: a primitive Dirichlet L-function $L(s, \chi)$ would have an Euler factor of the form $(1 - \chi(p) \cdot p^{-(s)})^{(-1)}$ at every prime including $p = 2$, with $\chi(2) \in \{0, \pm 1\}$ and similar character values elsewhere. The factor $(1 - 2^{(1-s)})$ is not of this form – the exponent shift $2^{(1-s)} = 2 \cdot 2^{(-s)}$ carries an explicit weight of 2 inside the local factor at $p = 2$ that no character can supply.

This is verifiable at the Dirichlet-series level. Compute the coefficients u_n of the series

$$\eta(s) \cdot (1 - 2^{(-s)}) = \sum_{n=1}^{\infty} u_n / n^s$$

term by term. The first sixteen coefficients are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
u_n	+1	-2	+1	0	+1	-2	+1	0	+1	-2	+1	0	+1	-2	+1	0

The pattern is periodic mod 4 with values $\{+1, -2, +1, 0\}$. The entry at $n \equiv 2 \pmod{4}$ is -2 , which exceeds the modulus of any complex unit: no Dirichlet character takes values outside $\{0\} \cup \{\text{roots of unity}\}$, so any candidate identification of M_g with a primitive-character L-function is ruled out at the coefficient level.

What survives is the Euler-product reading – and it is sharper than a character twist. Write the local-factor decomposition directly:

$$M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot (1 - 2 \cdot 2^{(-s)}) \cdot \prod_{p > 2} (1 - p^{(-s)})^{(-1)}.$$

At every odd prime p the local factor is ζ 's own inverse polynomial $(1 - p^{(-s)})^{(-1)}$. At $p = 2$ the local factor is the polynomial $(1 - 2 \cdot 2^{(-s)})$ itself – not the reciprocal of anything. The surgery at the bad prime therefore does more than substitute a character value: it inverts the analytic type of the local factor, replacing a pole-carrying inverse factor with a zero-carrying polynomial factor of degree one in $2^{(-s)}$. The coefficient 2 inside that polynomial is the same weight-2 obstruction that rules out the primitive-character identification at the Dirichlet-series level (the -2 entries in the $\{+1, -2, +1, 0\}$ coefficient pattern above): no Euler factor of the form $(1 - \chi(2) \cdot 2^{(-s)})^{(\pm 1)}$ with $\chi(2) \in \{0\} \cup \{\text{roots of unity}\}$ can produce it. The structure is formally analogous to the bad-prime local-factor modifications of L-function theory – M_g is not claimed to arise from an automorphic representation, and the installed factor (a degree-one polynomial in $2^{(-s)}$, not a reciprocal) is non-standard even by ramified-factor conventions. The phrase bad-prime modification of ζ is used throughout this paper in this descriptive sense.

The structural slogan for the document: the dome is ζ with surgically engineered $p = 2$ behavior – surgery that removes ζ 's local pole structure at 2 and installs a local zero structure with binary weight 2 in its place.

§4.3 Consequence: the binary theme is concrete

The narrow statement first, because it is pure algebra: the $c = 2$ doubling in the dome's denominator $\cosh(2x)$ is what produces the $p = 2$ local-factor surgery. The doubling forces the even-lattice exponential expansion of §3.2, and that lattice's Mellin image is exactly T13's surgered Euler product (§5.2 gives the family-level mechanism). This much is not metaphor; it is the algebraic content of M_g as an Euler product.

The framework's broader binary theme – Tower doubling in Paper 3, the binary cell $L = \ln 2$ in Paper 10, period-doubling via the Feigenbaum δ in the 400/11 residual of Papers 4/9 – is framework context, not algebra; its relation to this surgery is discussed where it belongs, in §7.4.

T13 reframes the rest of the document. Section §5 shows that the same $p = 2$ local-factor surgery is what selects the integer-producing branch in the natural Mellin-weight family; §6 shows that the visible arithmetic of $a(n)$ – Mersenne factors, Bernoulli factors, the von

Staudt-Clausen cancellation cascade – is the prime-by-prime shadow of T13’s Euler product.

§5 – The bifurcation theorem: η vs β routing in the dome family

Setup. Consider the small-integer family of weight functions

$$g_{\{a,b,c\}}(x) = \tanh^a(x) / \cosh^b(c \cdot x)$$

with a, b, c positive integers. The specific dome of §3 is the triple $(a, b, c) = (1, 1, 2)$. The question this section answers is structural: among the small triples, which ones produce integer normalized even-moment ratios $M_{\{a,b,c\}}(2n) / M_{\{a,b,c\}}(2)^n$? The answer is sharp.

§5.1 Theorem T8 (the bifurcation)

Theorem T8. Within the integer-coordinate cube $\{1 \leq a, b \leq 3, 1 \leq c \leq 3\}$, the normalized even-moment ratio $M_{\{a,b,c\}}(2n) / M_{\{a,b,c\}}(2)^n$ falls into three structural classes:

(i) η -branch – integers. The triple $(1, 1, 2)$ – the dome itself – is the unique member of the cube whose normalized ratio is integer-valued for all n . Its closed form is the sequence $a(n)$ of this paper.

(ii) β -branch – $(\pi/2)^{(n-1)}$ -scaled integers. The triple $(1, 1, 1) = \operatorname{sech}(x) \cdot \tanh(x)$ has normalized ratio

$$c_n = (2n - 1) \cdot |E_{\{2n-2\}}|$$

where $E_{\{2k\}}$ is the k -th Euler number (the secant numbers; OEIS A000364). These are integers, but they sit inside a generating function whose Mellin transform routes through Dirichlet $\beta(s)$ at odd integer arguments, producing a $(\pi/2)^{(n-1)}$ scaling relative to the dome’s η -routing.

(iii) Non-clean elsewhere. All remaining 25 triples of the 27-triple cube produce normalized ratios that are not integers and not clean $(\pi/2)$ -power-scaled integers; the algebraic structure of their Mellin spectra does not align with either η at even arguments or β at odd arguments. Eleven of these were classified in the original stress-test rounds – $(1, 2, 1)$, $(1, 2, 2)$, $(1, 3, 2)$, $(2, 1, 2)$, $(2, 2, 2)$, $(1, 1, 3)$, $(2, 1, 1)$, $(1, 3, 1)$, $(3, 1, 2)$, $(2, 2, 1)$, $(1, 2, 3)$ – and the remaining fourteen were classified in the full-cube sweep of 2026-06-10 (working notebook §2.13), completing the cube.

Verification status (Theorem-tier). Verified at numerical precision $\text{dps} = 40$ by a first independent architecture (Opus Chat Round 2, §1.8 working notebook), $\text{dps} = 50$ by a second independent architecture (Super Grok Round 2, §1.10) on five triples plus two control triples, and at exact arithmetic for the c_n closed form (Cowork, §1.6/§2.11). An attempted counter-example by a third architecture (Grok Heavy Round 2, §1.9) claimed integer behavior on three additional triples; on re-verification at $\text{dps} = 60$ all three were refuted, with one of them $(1, 1, 1)$ producing exactly the β -branch sibling sequence above $(3\pi/2, 25\pi^2/4, \dots)$ – confirming, not contradicting, the bifurcation. See §8 below for the methodological note. The full 27-triple cube was subsequently swept at $\text{dps} = 40$ (ratios $n = 1..5$) by a fourth architecture (Fable, working notebook §2.13): $(1, 1, 2)$ is the unique integer triple and $(1, 1, 1)$ the unique β -channel triple across the entire cube; all 25 others are non-clean. T8’s cube claim is therefore complete, not sampled. (Protocol: direct numerical Mellin quadrature at the stated dps ; a triple is classified η - or β -channel only if every tested ratio lies within $10^{(-25)}$ of the claimed exact value, non-clean otherwise. Logical structure of the uniqueness claim: a single ratio bounded away from every integer at the stated precision excludes a triple outright, so each of the 25 exclusions is rigorous and finitely witnessed – not sampled; the dome’s own integrality for all n is Theorem T15, §6.0. What remains numerical is only the positive η/β -channel identification of the two clean triples at untested orders.) In the publication fresh-eyes round of 2026-06-10 the full-cube sweep was independently reproduced (Uncle GPT, $n = 1..5$; working notebook §1.11), making the cube classification double-swept.

§5.2 The structural meaning of T8

T8 is a classification statement: the Mellin transform of a weight function in the dome family routes through one of two Dirichlet L-function channels at integer arguments, with sparse “non-clean” residue elsewhere. The two channels are:

- η -channel. Routes through Dirichlet $\eta(s)$ at even integer s . Produces (after natural ratio normalization) sequences with no residual π , hence integer-valued. The dome (1, 1, 2) sits here.
- β -channel. Routes through Dirichlet $\beta(s)$ at odd integer s . Produces integer-coefficient sequences scaled by $(\pi/2)^{(n-1)}$. The sibling $\text{sech} \cdot \tanh = (1, 1, 1)$ sits here.

The $c = 2$ doubling in the dome’s denominator $\cosh(2x)$ is what places (1, 1, 2) on the η -channel rather than the β -channel: the doubling forces the Dirichlet-series expansion of g into all-even support, where $\eta(s)$ operates, while the natural odd-symmetric $\tanh(x) \cdot \text{sech}(x)$ expansion places (1, 1, 1) on the all-odd $\beta(s)$ support. In Euler-product language (cf. T13): the $c = 2$ doubling is precisely what produces the $p = 2$ local-factor surgery that distinguishes the dome’s Mellin transform from a pure ζ Euler product or a pure β -character L-function.

T8 and T13 are the same fact viewed at two different levels. T13 says the dome’s Mellin transform is ζ -with-engineered- $p=2$ -surgery; T8 says this surgery is what selects the integer branch within the family. The integrality of $a(n)$ is therefore not the central phenomenon – the central phenomenon is the bifurcation of the family into η -routed and β -routed sub-families, with the dome’s specific $p = 2$ local-factor structure as the analytic mechanism behind the selection.

§5.3 The β -channel sibling sequence

The β -branch sibling closed form $c_n = (2n-1) \cdot |E_{\{2n-2\}}|$ is itself a clean result with the same flavor as $a(n)$ – both are integer-coefficient sequences from natural Mellin moments of natural hyperbolic weights – and provides a control comparison for the dome’s η -branch behavior. The first ten terms:

n	:	1	2	3	4	5	6	7	8	9	10
c_n	:	1	3	25	427	12465	555731	35135945	2990414715	329655706465	45692713833379

(Verified by Cowork at exact arithmetic via the Euler number identity $|E_{\{2k\}}| = A000364(k)$, $n = 1..10$. Values continue with growth proportional to $(2n)! / \pi^{(2n)}$, the same scaling as $a(n)$ up to the η vs β branch distinction.)

The standalone paper takes $a(n)$ as its primary object; the β sibling c_n is recorded here as the structural counterpart on the alternative branch, and as the most direct evidence for the bifurcation classification of T8.

§6 – The arithmetic shadow: prime structure of $a(n)$

The integer factorizations of $a(n)$ carry a clean prime-by-prime structure: every prime factor of $a(n)$ is exactly one of three species. This section presents that structure, verified at exact arithmetic for $n = 1..20$ (Working_Notes_Dome_Moment_Sequence.md §2.5–§2.9), as the prime-by-prime shadow of T13’s Euler product. Integrality itself is settled for all n by Theorem T15 below; the rest of the section describes which primes appear.

§6.0 Theorem T15 (integrality)

Theorem T15. For every $n \geq 1$, the normalized even-moment ratio

$$a(n) = (2^{(2n-1)} - 1) \cdot (2^{(2n)} - 1) \cdot |B_{\{2n\}}| \cdot 2^n / n$$

is an integer.

Proof. Write $|B_{2n}| = N_{2n} / D_{2n}$ in lowest terms. By the von Staudt–Clausen theorem, $D_{2n} = \prod_{\{(p-1) \mid 2n\}} p$, squarefree, and $2 \mid D_{2n}$ always (since $1 \mid 2n$); in particular N_{2n} is odd. It suffices to show, for every prime p ,

$$v_p(n \cdot D_{2n}) \leq v_p((2^{2n-1} - 1) \cdot (2^{2n} - 1) \cdot N_{2n} \cdot 2^n).$$

Case $p = 2$. The left side is $v_2(n) + 1$. On the right, both Mersenne-pair factors and N_{2n} are odd, so the right side is exactly n . The inequality $v_2(n) + 1 \leq n$ holds for all $n \geq 1$, since $2^{v_2(n)} \leq n$ gives $v_2(n) \leq \log_2 n \leq n - 1$. (The surplus $n - 1 - v_2(n)$ is precisely Theorem T3’s 2-adic valuation of $a(n)$.)

Case p odd with $(p - 1) \mid 2n$. The left side is $1 + v_p(n)$ (one factor of p from D_{2n} , plus the contribution of n). Let $d = \text{ord}_p(2)$. By Fermat’s little theorem $d \mid p - 1$, hence $d \mid 2n$, hence $p \mid 2^{2n} - 1$. By the lifting-the-exponent lemma for the odd prime p (applied to $2^{2n} - 1 = (2^d)^{(2n/d)} - 1$ with $p \mid 2^d - 1$):

$$v_p(2^{2n} - 1) = v_p(2^d - 1) + v_p(2n/d) \geq 1 + v_p(n),$$

where $v_p(2n/d) = v_p(n)$ because $d \mid p - 1$ forces $p \nmid d$, and p is odd. The single Mersenne factor $(2^{2n} - 1)$ therefore absorbs both the von Staudt–Clausen denominator’s p and the full p -content of $1/n$. (Theorem T9’s denominator compounding for composite n is the $v_p(n) \geq 1$ instance of this case; the $n = 3$ worked example is $d = 2$, $v_3(2^6 - 1) = v_3(3) + v_3(3) = 2$.)

Case p odd with $(p - 1) \nmid 2n$ and $p \mid n$. The left side is $v_p(n)$ (no von Staudt–Clausen contribution). By the p -integrality of $B_{2n}/2n$ when $(p - 1) \nmid 2n$ – recorded by Adams (1878, p. 269) in divisibility form, explicitly as an observation he could not prove (“I have not succeeded, however, in obtaining a general proof of this proposition, though I have no doubt of its truth”); the divisibility form was first proved by Voronoï in 1889, as a student under Markov (Syta & van de Weygaert 2009 record the three-week proof as his research debut); the full p -integral statement is the modern treatment, Ireland & Rosen, Prop. 15.2.4 – we have $v_p(B_{2n}) \geq v_p(2n) = v_p(n)$. Since $p \nmid D_{2n}$ in this case, the numerator carries $v_p(N_{2n}) \geq v_p(n)$, absorbing the $1/n$.

Case p odd with $(p - 1) \nmid 2n$ and $p \nmid n$. The left side is 0; nothing to absorb.

All four cases close, so $n \cdot D_{2n}$ divides the numerator product and $a(n) \in \mathbb{Z}$. ■

Remark (T5 is Adams’ 1878 observation). The third case identifies the mechanism behind Theorem T5’s cancellation table: the Bernoulli-source primes cancelled by $1/n$ – 11 at $n = 11$, 13 at $n = 13$, 7 at $n = 14$, 5 at $n = 15$, 17 at $n = 17$, 19 at $n = 19$ – are exactly the Adams primes of their rows, the primes $p \mid n$ with $(p - 1) \nmid 2n$ whose presence in the numerator of B_{2n} is guaranteed by Adams’ observation (in its proved modern form) and whose removal by the explicit $1/n$ is exact. T5 is the table-level shadow of a pattern Adams recorded in 1878 – and explicitly could not prove. The citation anchors §6 to the classical Bernoulli literature, and carries a methodological bonus: Adams tier-marked his own claims in the cited pages – “I have proved that...” for the prime-index divisibility, “I have also observed that... I have no doubt of its truth” for the proposition used here. The theorem/observation distinction, practiced by hand, in 1878.

Integrality is verified at exact arithmetic for $n = 1..30$ (Fable Chat fresh-eyes pass, 2026-06-10, sympy exact rationals; independently re-verified by Fable Cowork the same day), extending the $n = 1..10$ and $n = 1..20$ ranges of the earlier rounds, and extended to $n = 1..60$ in the pre-deposit round (Fable Chat, 2026-06-12); Theorem T15 settles all n . (Proof contributed by Fable Chat, publication fresh-eyes round 2026-06-10.)

§6.1 Theorem T4 (three-species prime partition)

Theorem T4. For all $n \geq 1$, each prime p dividing $a(n)$ is exactly one of:

- (I) $p = 2$. The 2-power content of $a(n)$ is given by $v_2(a(n)) = n - 1 - v_2(n)$ for $n \geq 1$ (Theorem T3 below).

(II) Mersenne / Fermat primes inherited from the doubling pair. Every prime appearing in either of the factors $(2^{(2n-1)} - 1)$ or $(2^{(2n)} - 1)$ is, by definition, a prime that divides $2^k - 1$ for some k . These are candidate Mersenne primes when the exponent $2n-1$ is prime (primality of the exponent is necessary, not sufficient: $2^{11} - 1 = 2047 = 23 \cdot 89$ at $n = 6$) together with their composite extensions – primes p for which the multiplicative order of 2 modulo p divides $2n-1$ or $2n$.

(III) Bernoulli numerator primes from $|B_{\{2n\}}|$. Every prime p in the numerator of $B_{\{2n\}}$ appears in $a(n)$ unless cancelled. The classical irregular primes appear in $a(n)$ at the n corresponding to their B index: 691 ($\in B_{12}$), 3617 ($\in B_{16}$), 43867 ($\in B_{18}$), 283 and 617 ($\in B_{20}$), 131 and 593 ($\in B_{22}$), 103 and 2294797 ($\in B_{24}$), and so on. For $n \geq 10$ the Bernoulli numerators are generally composite – e.g. $174611 = 283 \cdot 617$, $854513 = 11 \cdot 131 \cdot 593$ – and it is their irregular prime factors that appear; the small regular factors (11, 13, 7, 5, 17, 19, ...) are Adams primes, removed exactly by the $1/n$ mechanism when $p \mid n$ (Theorems T15 and T5).

Mechanism. The three-species partition is enforced by the von Staudt–Clausen theorem on Bernoulli denominators. By von Staudt–Clausen, the denominator of $B_{\{2n\}}$ (in lowest terms) is exactly $\prod_{\{(p-1) \mid 2n\}} p$. Every such prime divides $2^{(2n)} - 1$: by Fermat’s little theorem, $\text{ord}_p(2)$ divides $p - 1$, which divides $2n$. The Mersenne factor $(2^{(2n)} - 1)$ therefore absorbs the entire Bernoulli denominator, and the cancellation is exact and complete: no rational-arithmetic primes survive in $a(n)$ other than those in species (I)–(III). (Sentence repaired 2026-06-12 – the previous wording was garbled; caught by Fable Chat in the pre-deposit round.)

§6.2 Theorem T3 (2-adic valuation)

Theorem T3. For all $n \geq 1$,

$$v_2(a(n)) = n - 1 - v_2(n)$$

where $v_2(m)$ is the standard 2-adic valuation of m (the exponent of 2 in m ’s factorization, with $v_2(1) = 0$). Verified at exact arithmetic for $n = 1..20$.

Proof sketch. From T1, the 2-power content of $a(n)$ tracks (a) the 2-power factor $2^n / n$ directly, contributing $n - v_2(n)$; (b) the 2-power content of $(2^{(2n-1)} - 1)$, which is 0 since $2^{(2n-1)} - 1$ is odd; (c) the 2-power content of $(2^{(2n)} - 1)$, which is 0 by the same argument; (d) the 2-power content of $|B_{\{2n\}}|$, which by the von Staudt–Clausen mechanism is exactly -1 (Bernoulli numerators are odd; the denominator carries exactly one factor of 2). Summing: $v_2(a(n)) = (n - v_2(n)) + 0 + 0 - 1 = n - 1 - v_2(n)$. ■

§6.3 Theorem T5 (the $/n$ cancellation rule)

Theorem T5. A Bernoulli numerator prime p in $|B_{\{2n\}}|$ appears in $a(n)$ unless p divides n . When p divides n , the explicit $1/n$ cancels one factor of p ; when p^2 divides n , additional cancellation may occur via the doubling-pair factors.

Verified at exact arithmetic for $n = 6..20$ against the explicit factorizations of $B_{\{2n\}}$ numerator and $a(n)$. Representative rows:

n	$B_{\{2n\}}$ numerator primes	$a(n)$ carries Bernoulli primes?	Cancellation mechanism
6	691	691 retained	no cancellation (gcd(691, 6) = 1)
8	3617	3617 retained	no cancellation (gcd(3617, 8) = 1)
9	43867	43867 retained	no cancellation

10	283, 617	both retained	no cancellation
11	11, 131, 593	131 and 593 retained; 11 cancelled	11 cancelled by $1/n = 1/11$
12	103, 2294797	both retained	no cancellation
13	13, 657931	657931 retained; 13 cancelled	13 cancelled by $1/n = 1/13$
14	7, 9349, 362903	9349 and 362903 retained; Bernoulli-source 7 cancelled†	7-from- $B_{\{28\}}$ cancelled by $1/n = 1/14$
15	5, 1721, 1001259881	1721 and 1001259881 retained; 5 cancelled	5 cancelled by $1/n = 1/15$
16	37, 683, 305065927	all three retained	no cancellation
17	17, 151628697551	151628697551 retained; 17 cancelled	17 cancelled by $1/n = 1/17$
18	26315271553053477373	retained (single giant irregular prime)	no cancellation
19	19, 154210205991661	154210205991661 retained; 19 cancelled	19 cancelled by $1/n = 1/19$
20	137616929, 1897170067619	both retained	no cancellation

† Note on $n = 14$: the prime 7 does appear in $a(14)$, but its source is the Mersenne factor $2^{\{27\}} - 1 = 7 \cdot 73 \cdot 262657$, not $B_{\{28\}}$. The Bernoulli-numerator contribution of 7 is the one cancelled by $1/n = 1/14$; the Mersenne-factor 7 is independent. T5 is a statement about Bernoulli-source primes specifically – the three-species partition T4 keeps each species's appearance bookkeeping separate.

(The cancellation table is mechanical through $n = 20$ and continues to higher n by the same rule. Minor cases involving prime-power-in- n behavior unfold by T9 below.)

§6.4 Theorem T9 (denominator compounding for composite n)

Theorem T9. When n has a prime-power factor p^k with $k \geq 2$, the Mersenne factor $(2^{(2n)} - 1)$ may carry p^j with $j \geq k$; clearing this requires both the von Staudt-Clausen denominator's p contribution and the explicit $1/n$'s p contribution, and possibly multiple invocations.

Worked example, $n = 3$. $(2^6 - 1) = 63 = 3^2 \cdot 7$. By T4, both factors of 3 must be cancelled. The first 3 is cancelled by the denominator of $B_6 = 1/42$ (denom $42 = 2 \cdot 3 \cdot 7$ carries one factor of 3, by von Staudt-Clausen with $(3-1) \mid 6$). The second 3 is cancelled by $1/n = 1/3$. Net: $a(3) = (2^5-1) \cdot (2^6-1) \cdot |B_6| \cdot 2^3 / 3 = 31 \cdot 63 \cdot (1/42) \cdot 8 / 3 = 31 \cdot 63 \cdot 8 / 126 = 124$. The 3-content has been fully cleared; the $124 = 4 \cdot 31$ carries only the Mersenne 31 and the 2-content from T3 ($v_2(124) = 2 = 3 - 1 - v_2(3) = 3 - 1 - 0 = 2$). ✓

Note on Q9 (open question). A general prime-by-prime characterization $v_p(a(n))$ for arbitrary primes p generalizing T3 – the odd-prime valuation atlas of $a(n)$ – is a natural research target. The expected four-term decomposition involves the multiplicative order of 2 mod p (governing the Mersenne contributions via the lifting-the-exponent lemma), the Bernoulli structure (governing the $|B_{\{2n\}}|$ contribution), the explicit $1/n$ (cancelling primes dividing n), and the von Staudt-Clausen interactions. See §9 below.

§6.5 Growth (T6, T7)

Theorem T6. The leading-order ratio behavior is $a(n+1) / a(n) \sim (32/\pi^2) \cdot n^2$, with $32/\pi^2 \approx 3.242\dots$.

Theorem T7. The leading-order asymptotic is $a(n) \sim 8^n \cdot (2n)! / (n \cdot \pi^{(2n)})$.

Both follow from the closed forms of §2 together with the Stirling approximation and standard estimates for $|B_{\{2n\}}|$. The asymptotics make the integer character of $a(n)$ all the more striking: the leading order carries $\pi^{(2n)}$, Stirling factorials, and exponentials, yet the result is an integer; the cancellation is exact and structurally enforced (T4, T5).

§7 – Framework context

This section places $a(n)$ in the broader Gap Geometry framework – briefly, since the standalone is mathematically self-contained without it. Readers interested only in the number-theoretic content may skip directly to §8.

§7.1 The framework constant K_{AUD}

The framework’s central constant is

$$K_{\text{AUD}} = \sqrt{2} \cdot \ln 2 \approx 0.980258\dots$$

derived independently in three places: (a) Paper 3 as a geometric $\times$ binary intersection involving the H_4 polytope; (b) Paper 10 as the saturation coordinate of Marsaglia’s Algorithm EA, falling out of the quadratic $b^2 - 4b + 2 = 0$ at the binary cell $L = \ln 2$; (c) Paper 11 as the central constant of the η corridor in 3D Wilson-Fisher critical phenomena. The same constant arrives via three structurally distinct derivation chains; this is the framework’s K_{AUD} ubiquity signature.

§7.2 The dome as carrier of K_{AUD} ’s two ingredients

The dome $g(x) = \tanh(x) / \cosh(2x)$ encodes both factors of $K_{\text{AUD}} = \sqrt{2} \cdot \ln 2$ independently:

- The $\sqrt{2}$ ingredient appears as the self-reference fixed point of the dome at $u = \sqrt{2}$ (Paper 11 Theorem 9.4). See Paper 11 for the precise algebraic statement; the present paper takes the fixed-point existence as a referenced result.
- The $\ln 2$ ingredient appears as the dome integral (Paper 11 Theorem 9.5): $\int_0^\infty g(x) dx = (\ln 2) / 2$.

Their product is K_{AUD} itself. The dome is the analytic carrier of the framework’s central constant – both ingredients algebraically visible inside one weight function. The doubling structure is moreover summable: by the alternative definition $g(x) = \tanh(2x) - \tanh(x)$ (§3.1), the dome evaluated along the Binary Tower’s rungs telescopes exactly, $\sum_{k \geq 0} g(2^k x) = 1 - \tanh(x)$ – the Tower is the literal summation skeleton of the dome, and the dome is the per-rung increment of saturation.

§7.3 The sequence as arithmetic shadow

The Mersenne factors $(2^{(2n-1)} - 1)(2^{(2n)} - 1)$ are the discrete arithmetic shadow of the dome’s binary doubling structure (the $c = 2$ in $\cosh(2x)$ and its analytic consequences); the Bernoulli factor $|B_{\{2n\}}| \cdot 2^n / n$ is the arithmetic shadow of the dome’s analytic ζ -content (via $\eta = (1 - 2^{(1-s)})\zeta$ and the standard $\zeta(2n) \leftrightarrow B_{\{2n\}}$ bridge). The sequence $a(n)$ is therefore the number-theoretic carrier of the same structural object that Paper 10 derives algorithmically and Paper 3 derives geometrically – a fourth, arithmetic path to the same K_{AUD} -organized structural object.

§7.4 K_{AUD} as fixed-point coordinate of binary balancing

A useful reframing surfaced in cross-framework discussion: K_{AUD} is best understood not as the framework’s central object but as a fixed-point coordinate of an underlying binary balancing

operation. Paper 10's $b^2 - 4b + 2 = 0$ derivation makes this transparent – K_{AUD} is literally the fixed point of the binary-cell endpoint-balancing equation. The same is true in Paper 3's $H_4 \times$ binary intersection and in Paper 11's η corridor saturation. T13's $p = 2$ local-factor surgery is the same phenomenon at the Mellin level: ζ 's local Euler factor at $p = 2$ is replaced by a different factor, and the dome is what survives that replacement under the natural even-moment-ratio normalization.

The framework's "binary" theme – Tower doubling in Paper 3, the binary cell $L = \ln 2$ in Paper 10, the dome's $c = 2$, period-doubling via the Feigenbaum δ in the 400/11 residual of Papers 4/9 – viewed this way, is a renormalization operation – a scale-doubling fixed-point selection – rather than a Z_2 -arithmetic operation. The next-abstraction-up mathematical home is a category of scale-doubling transformations and their fixed points; the dome and $a(n)$ are particular objects in that category. The L-function classification of weight families by Mellin-image branch (η vs β) is the categorical statement that the sequence belongs to one specific sub-object – the η -routed one – and that membership is the structural fact that produces integer arithmetic.

The standalone does not pursue this categorical formulation further; §9 lists it as a concrete open research target.

§8 – Independent reconstruction notes

This section records the multi-architecture independent-reconstruction trace that underlies the verification status of the results ledger above (Theorems T1–T9, T13, T15; Observation O10; Readings R11–R12; Decision D14). The discipline is transparency-of-provenance: each Theorem-tier claim is cross-architecture verified, each Conjecture-tier claim is explicitly marked, and one informative disagreement is recorded as it makes the result stronger rather than weaker.

A note on proportion, and on stopping. The pre-deposit rounds of 2026-06-10 through 2026-06-12 recorded in the footer were primarily editorial verification – labels, wording, counts, conversion-layer integrity – conducted with eyes open for mathematical error and finding the mathematics already settled. The substance carries a longer ledger: this paper's working notebook alone records thirty-seven dated trace and verification entries, and the dome line of work – begun in April 2026 under earlier working names and developed through Paper 11 and its Verification Companion – carries dozens more across every stage, each recorded at the time. Verification before deposit is also deliberately bounded: past a point, further passes test the testers rather than the theorems. The sequence and its proofs are preserved complete in this single document; testing continues after publication, by readers – the verification-first infrastructure exists for exactly that.

§8.1 Architectures and roles

Nine language-model architectures participated in the reconstruction. Named by working name as they appear in the project's internal notebooks. (This list covers the mathematical reconstruction and publication fresh-eyes rounds through 2026-06-11; the 2026-06-12 pre-deposit editorial review – Haiku, Sonnet, Opus, Fable Chat – and a context-free cold-read audit are recorded in the footer trail and in the working notebook, §1.13–§1.17.)

- Grok Heavy. Performed two distinct passes. In the memory-blind round (given the eight integers only, no framework context, no theorem statement) it independently reverse-engineered the cyclotomic $\Phi_d(2)$ decomposition from data and surgically identified 691 and 3617 as primitive divisors from a non-Mersenne source – without naming them as Bernoulli numerator primes, as the memory-blind run had no context lens for that. In the pattern-search round (given the framework artifacts) it produced O10 (the K_{AUD} ubiquity bridge, Observation-tier, connecting the sequence to Paper 3's $H_4 \times$ binary intersection and Paper 10's Algorithm EA saturation). In Round 2 of T8 verification it attempted three counter-examples at numerical precision $dps = 40$, all three of which were refuted at exact arithmetic – the contested-and-resolved trace of §8.3. In the publication fresh-eyes round (2026-06-10), with its agent team, verified the factorizations of $a(6)$ and $a(11)$ at exact

- arithmetic (independently confirmed; working notebook §2.15a) and triaged the §9 open questions – the Q10 prioritization in that message is what triggered the same-day closure of the §9.2 table, and the Q12-as-unification-layer reading is recorded alongside it.
- Uncle GPT. Given the eight integers plus Theorem 9.7 of Paper 11, crystallized the closed form into the correct normalization (the ratio $a(n) = M(2n)/M(2)^n$) and supplied the structural-completion pass that produced four reframings: K_{AUD} -as-fixed-point-coordinate, η -branch-as-protagonist, “binary”-as-renormalization (not Z_2 -arithmetic), and the sequence as a compatibility object between cyclotomic and Mellin arithmetic worlds. In the L-function classification follow-up, supplied T13’s η -to- ζ rewrite (the document’s central algebraic identity) and the bad-prime-modification linguistic framing – the load-bearing structural contribution to the paper. In the publication fresh-eyes round (2026-06-10), independently reproduced the full 27-triple cube sweep at $n = 1.5$ and supplied the verified-vs-proved protocol discipline now stated in §5.1.
 - Super Grok. Supplied T2 (the A002105 $\times$ Mersenne-companion factorization) – the document’s reader-friendly entry point and the headline closed form. Independently re-derived in Round 2 the bifurcation theorem at numerical precision $\text{dps} = 50$, including two control triples beyond the prompt, confirming T8’s classification across the triples tested in that round (the full 27-triple cube was closed later; see §5.1).
 - Opus Chat. In Round 2, produced T8 (the bifurcation theorem): the dome family’s small-triple integer-coordinate cube splits into η -channel integer producers, β -channel $(\pi/2)^{(n-1)}$ -scaled integer producers, and non-clean elsewhere; the dome (1, 1, 2) is the unique η -channel member within the cube. The bifurcation classification reorganizes the entire document at D14.
 - Gemini (3.1 Pro and 3.5 Thinking, publication fresh-eyes round, 2026-06-10). Independent spot verifications on the post-fix draft: T2 arithmetic re-derived by hand for $n = 1.4$ (3.1 Pro); direct numerical quadrature reproducing $a(1)$ – $a(4)$ as exact integers, the T13 closed form at $k = 1.4$, and the (1, 1, 1) β -channel ratios $3\pi/2$ and $25\pi^2/4$ to 30 digits (3.5 Thinking). Both passes independently confirmed that the §8.3 contested-and-resolved trace reads as methodological strength rather than weakness.
 - Opus Cowork (this writeup). Performed the central exact-arithmetic verification pass at $\text{dps} \geq 60$ on all Theorem-tier claims; supplied the closed-form derivation of the β -channel sibling sequence $c_n = (2n-1) \cdot |E_{2n-2}|$; performed the Dirichlet-series-coefficient-level verification of T13’s “not a primitive Dirichlet character” reading via the mod-4 pattern $\{+1, -2, +1, 0\}$; derived T3 (the 2-adic valuation rule $v_2(a(n)) = n - 1 - v_2(n)$) and the three-species partition mechanism for T4 via von Staudt–Clausen. Assembled this writeup, including the reorganization of the document’s spine around T13/D14 after the L-function classification round.
 - Sonnet Chat (v1 fresh-eyes review). Caught the η -notation collision between Paper 11’s anomalous-dimension η and this paper’s Dirichlet η on first-read of v1; the disambiguation note in §3.2 addresses readers arriving from the corridor paper before they hit “ η -branch” language and try to import critical-phenomena meaning into it. Also confirmed the structural choices of v1: that the “bad-prime modification of ζ at $p = 2$ ” framing is the right central narrative (sharper than “integer sequence with structure”), that the closing slogan earns itself, and that the §9.5 scope discipline (“what the document does not claim”) is correctly placed.
 - Fable (Cowork, from 2026-06-10). Closed the 27-triple cube with the first full sweep ($\text{dps} = 40$, $n = 1.5$; working notebook §2.13); corrected the §3.1 decay asymptotics and supplied the exact exponential expansion now in §3.2; integrated the publication fresh-eyes round and independently re-verified every patch claim at exact arithmetic (T15 at $n = 1.30$, the §4.2 discrepancy demonstration, the Q10 row quadrature).
 - Fable Chat (publication fresh-eyes round, 2026-06-10). Caught the §4.2 Euler-product template error (the former f_2 display; resolved by the polynomial local-factor presentation now in §4.2); contributed Theorem T15’s complete integrality proof (von Staudt–Clausen + lifting-the-exponent + Adams) and the identification of T5’s cancellation rule as Adams’ theorem; closed the Q10 $\tanh \cdot \text{sech}$ row. In the same-day hostile audit of the §9.2 closed forms: independently re-derived both forms by blind partial fractions, dissolved the (1, 2, 2) persistence question (the coefficient law is an algebraic identity, so no verification horizon exists), supplied the explicit (1, 2, 2) closed form, and surfaced the dome’s alternative definition $g = \tanh(2x) - \tanh(x)$ with its doubling-ladder telescoping sum (§3.1). All contributions proposed as patch blocks under single-writer discipline; integrated and re-verified by Fable (Cowork).

§8.2 What converged

The following claims were independently arrived at by two or more architectures, with full cross-verification at exact arithmetic by Opus Cowork:

- T1 (Bernoulli closed form): Uncle GPT + Super Grok + Opus Cowork
- T2 ($A002105 \times$ Mersenne factorization): Super Grok, verified by Opus Cowork and re-derived from T1
- T3 (2-adic valuation rule): Opus Cowork, independently re-derived by Uncle GPT
- T4 (three-species prime partition): Opus Chat identified the species structure; Opus Cowork derived the von Staudt–Clausen mechanism; Uncle GPT unified the cancellation principle into the “denominator-structure” framing
- T5 (the $/n$ cancellation rule): Opus Chat identified two cases; Opus Cowork extended to eight verified cases through $n = 20$; Uncle GPT confirmed
- T6, T7 (growth asymptotics): Opus Cowork, corrected after a first-pass slip by Grok Heavy
- T8 (bifurcation η vs β): Opus Chat (Round 2) + Super Grok (Round 2) + Opus Cowork (β -sibling closed form at exact arithmetic)
- 010 (K_AUD ubiquity bridge; Observation): Grok Heavy pattern-search pass, mapped at the framework level by Opus Cowork
- R11, R12 (four structural reframings + closing categorical proposal; Readings): Uncle GPT, accepted as document-reshaping pass
- T13 (η -to- ζ rewrite with $p = 2$ local-factor surgery): Uncle GPT, verified algebraically and at the Dirichlet-series coefficient level by Opus Cowork
- D14 (η vs β as central dichotomy; Decision): Uncle GPT, accepted as document-reshaping decision

§8.3 One informative disagreement, recorded transparently

In the second round of stress-testing the bifurcation theorem T8, Grok Heavy claimed (at $dps = 40$) that three additional triples within the dome family produced integer normalized even-moment ratios: (1, 1, 1), (1, 2, 2), and (1, 3, 2). The claim contradicted Opus Chat’s independent derivation that, within the cube, only (1, 1, 2) produced integers.

The contradiction was resolved by Opus Cowork at $dps = 60$ on exact-arithmetic recomputation. All three claimed counter-examples failed: (1, 1, 1) produces $3\pi/2$, $25\pi^2/4$, $427\pi^3/8$, ... – exactly the β -channel sibling pattern $(\pi/2)^{(n-1)} \cdot c_n$ with $c_n = (2n-1) \cdot |E_{2n-2}|$ identified independently – while (1, 2, 2) and (1, 3, 2) produce non-clean numerical residuals matching neither the η -channel nor the β -channel form. Grok Heavy had reported (1, 1, 1) as producing “1, 3, 15, 105, 945” – the double-factorial pattern $(2n-1)!!$ – substituting a recognized famous sequence for his own quadrature output: the actual ratio at $n = 2$ is $3\pi/2 \approx 4.71$, which rounds to 5, not 3, so no rounding path explains the report (working notebook §1.9). He committed to the pattern-matched identification rather than reporting the residual π content of the numerical quadrature.

The post-hoc diagnostic is that pattern-matching dominated computation: Grok Heavy’s recognition of a famous sequence took precedence over its own numerical evidence. The contradiction was informative – it exposed that (1, 1, 1) is the β -channel sibling, and this fact strengthened T8’s structural reading. The bifurcation theorem is more trustworthy with the contested-and-resolved trace on record than it would have been from uniform consensus. Grok Heavy’s pattern-recognition strength is, separately, the same strength that produced 010’s K_AUD ubiquity bridge – the slip mode and the productive mode are two faces of the same architecture-level disposition.

This observation is recorded as a methodological note: a consensus answer is less trustworthy than a contested-and-resolved one – the disagreement made T8 stronger. The full traces, including precise dps values, exact-arithmetic counter-computations, and the cross-architecture timing of each derivation, live in the project’s internal working notebook (Working_Notes_Dome_Moment_Sequence.md §1.1–§1.10 and Cross_Framework_Connections_Notes.md §A.1–§A.3).

§8.4 What this round did not produce

For completeness, the cross-framework stress-test rounds did not surface:

- A primitive Dirichlet character identification for M_g (correctly: T13 shows there is none – M_g is a bad-prime modification of ζ , not a character L-function).
- A modular-form interpretation of the dome's Mellin spectrum, despite the irregular primes 691 and 3617 appearing in $a(n)$ at exactly the positions where they govern weight-12 and weight-16 modular form congruences. The connection is plausible (Q11 below) but no architecture surfaced it concretely in the stress-test rounds.
- A general odd-prime valuation atlas for $v_p(a(n))$ (Q9).
- The categorical formulation of the η -vs- β classification (Q12).

These remain as open structural targets – the next mathematical research directions surfaced by the round.

§9 – Open structural questions

The structural reading of §4–§5 leaves four open research directions, each well-posed enough to be a concrete entry point into existing research literature on Mellin transforms, L-function classification, and modular forms.

§9.1 Q9 – Odd-prime valuation atlas

Generalize T3 from the 2-adic case to a prime-by-prime characterization $v_p(a(n))$ for arbitrary odd primes p . The expected four-term decomposition is:

$$v_p(a(n)) = v_p[(2^{2n-1} - 1)] + v_p[(2^{2n} - 1)] + v_p(|B_{2n}|) - v_p(n) + (\text{von Staudt-Clausen interactions})$$

The first two terms are governed by the multiplicative order $\text{ord}_p(2)$ via the lifting-the-exponent lemma. The third term is governed by Bernoulli numerator/denominator structure, hence by Kummer's irregularity criterion. The fourth term is the explicit $1/n$ cancellation. The interactions, recorded in T9 above, govern the cases when n carries higher prime powers. A complete atlas would produce a prime-by-prime arithmetic map of the sequence.

§9.2 Q10 – Mellin-to-L-function table for the dome family

The bifurcation theorem T8 raises a classification question: which weight functions in the dome family $g_{\{a,b,c\}}(x) = \tanh^a(x) / \cosh^b(c \cdot x)$ map under the Mellin transform into which Dirichlet L-function branch? The table – all six rows now established (the final two closed in the 2026-06-10 follow-up; working notebook §2.15) – is:

Weight	Mellin image at integer s	L-function family
$\text{sech}(x)$	$2 \Gamma(s) \beta(s)$	β at odd integers
$\text{sech}(2x)$	$2^{1-s} \Gamma(s) \beta(s)$	β at odd integers (rescaled)
$\tanh(x) \cdot \text{sech}(x)$	$2 \Gamma(s) \beta(s-1)$	β at odd integers (shifted argument)
$\tanh(x) \cdot \text{sech}(2x)$ [the dome]	$\Gamma(s) \cdot \eta(s) \cdot (2^{s-1} / 2^{2s-1})$	η at even integers
$\tanh(x) \cdot \text{sech}^2(2x)$	$\Gamma(s) \cdot 2^{1-s} \cdot [\beta(s-1) - (1 - 2^{-s}) \cdot \eta(s)]$	mixed: shifted β + the dome's η -block (established)
$\tanh^2(x) \cdot \text{sech}(2x)$	$\Gamma(s) \cdot [2^{2-s} \eta(s-1) - 2^{1-s} \beta(s)]$	mixed $\eta(s-1) + \beta(s)$ (established,

(The dome row is written in its η -form; by Theorem T13 it equals $\Gamma(s) \cdot 2^{(1-s)} \cdot \zeta(s) \cdot (1 - 2^{(-s)})(1 - 2^{(1-s)})$ identically – the two representations are the same function.)

The $\tanh \cdot \operatorname{sech}$ row follows from $\tanh \cdot \operatorname{sech} = (-\operatorname{sech})'$ and one integration by parts: $M(s) = (s-1) \cdot 2\Gamma(s-1)\beta(s-1) = 2\Gamma(s)\beta(s-1)$, verified numerically at $s = 2, 4, 6$; it reproduces $c_n = (2n-1) \cdot |E_{2n-2}|$ with the $(\pi/2)^{(n-1)}$ scaling exactly. The shifted argument – β at odd arguments when $s = 2n$ is even – is the cleanest one-line statement of why the $(1, 1, 1)$ sibling sits on the β -channel. The last two rows were closed on 2026-06-10 and hostile-audited the same day (working notebook §2.15–§2.16): both coefficient laws are proved for all indices by exact partial fractions (the decompositions are algebraic identities, so no verification horizon exists), and both closed forms are confirmed by quadrature at six values of s each, including non-integer s , to $\leq 10^{(-51)}$. Special values: $\int_0^\infty \tanh^2 x \cdot \operatorname{sech} 2x \, dx = 1 - \pi/4$ and $\int_0^\infty \tanh x \cdot \operatorname{sech}^2 2x \, dx = (1 - \ln 2)/2$, both exact. Three conventions are load-bearing: the linear-coefficient Dirichlet series have abscissa of convergence $\operatorname{Re}(s) = 2$, so at small s the closed forms are their analytic continuations (quadrature confirms the continued values directly); the coefficient periodicity is mod 4 on the $e^{(-2x)}$ lattice used in §3.2 and mod 8 on the $e^{(-x)}$ lattice; and $\eta(0) = \beta(0) = 1/2$ enter the $s = 1$ values.

Two structural remarks. First, the η -component of the $(1, 2, 2)$ image, $(1 - 2^{(-s)}) \cdot \eta(s)$, is identical to the dome's own surgered ζ -content from T13 – the $(1, 2, 2)$ member is not vaguely “mixed” but exactly one shifted β -block plus the dome's η -block. Second, this is no accident, and it takes two lines. Line one: $\tanh(2x) - \tanh(x) = \tanh(x) \cdot \operatorname{sech}(2x)$ – which is the dome itself (§3.1). Line two: multiply through by $\operatorname{sech}(2x)$ to get $\tanh(x) \cdot \operatorname{sech}^2(2x) = \tanh(2x) \cdot \operatorname{sech}(2x) - \tanh(x) \cdot \operatorname{sech}(2x)$, whence at the Mellin level $M_{\{(1,2,2)\}}(s) = 2^{(-s)} \cdot M_{\{(1,1,1)\}}(s) - M_{\{(1,1,2)\}}(s)$ – the $(1, 2, 2)$ member is literally the dilated β -sibling minus the dome. The family mixes out of recognizable pieces.

The mechanism that closes the table is general. Every member of the family is a rational function of $q = e^{(-x)}$ with cyclotomic denominators. Partial fractions then force the exponential-expansion coefficients to be quasi-polynomial – a polynomial in the index times a periodic character – and termwise Mellin integration turns any quasi-polynomial coefficient law into a finite combination of Dirichlet L-functions at shifted arguments (the polynomial degree sets the maximal shift; the period sets the character modulus). Pole multiplicity at the cyclotomic points is what buys polynomial coefficient growth, hence shifts – a mechanism with two instances now proved end-to-end: the $(2, 1, 2)$ double pole at $q = -1$ produces exactly the $\eta(s-1)$ shift, and the $(1, 2, 2)$ double pole at $q = \pm i$ produces exactly the $\beta(s-1)$ shift. In this light the dome's uniqueness sharpens: $(1, 1, 2)$ is the unique member of the cube whose Mellin image collapses to a single, unshifted Dirichlet object (η) with bounded coefficients – simple poles, one lattice, no shift. Its sibling $(1, 1, 1)$ pays one shift ($\beta(s-1)$); every other member pays in shifts and mixtures. This makes precise both the “convolution destroys channel purity” observation and the denominator-structure intuition recorded in §1.9 of the working notebook. What remains open in Q10 is the general pole-multiplicity-to-shift dictionary for arbitrary (a, b, c) – the table for the natural small family is settled.

§9.3 Q11 – Eisenstein-series / modular-form interpretation

The appearance of irregular Bernoulli primes in $a(n)$ at exactly the positions where they govern modular-form congruences is suggestive:

- $a(6)$ contains 691, the irregular prime of B_{12} , which governs Ramanujan's $\tau(p) \equiv 1 + p^{11} \pmod{691}$ congruence between the weight-12 cusp form Δ and the weight-12 Eisenstein series E_{12} .
- $a(8)$ contains 3617, the irregular prime of B_{16} , which governs the corresponding weight-16 congruence.
- $a(9)$ contains 43867 $\in B_{18}$; $a(10)$ contains the irregular primes of B_{20} ; and so on for all higher n where B_{2n} carries irregular numerator primes.

Conjecture: the dome $g(x)$ admits an interpretation as the Mellin pre-image of a specific

Eisenstein-series combination, perhaps at varying weight $2n$, with the irregular-prime content of $a(n)$ the visible signature of weight- $2n$ Eisenstein-cusp form congruences. The conjecture connects the sequence's number-theoretic content to a deep area of automorphic-forms theory.

The closest classical entry points are Berndt's exposition of Ramanujan's notebooks on Bernoulli congruences, Knopp's work on modular forms with Eisenstein-series Mellin representations, and Serre's Cours d'arithmétique on the analytic properties of Eisenstein series.

§9.4 Q12 – Categorical formulation

Make precise the category of weight functions and Mellin transforms within which T8 is a classification statement. Concrete proposal:

- Objects: pairs (g, M_g) where g is a hyperbolic weight function and M_g is its Mellin transform.
- Morphisms: scale/dilation operations $g(x) \mapsto g(cx)$, multiplication by elementary hyperbolic factors, convolution-type operations preserving the Dirichlet-series structure of M_g .
- Functor: the Mellin transform $M : g \mapsto M_g$.

The classification statement of T8 becomes a categorical statement: the η -image and β -image form distinguished sub-categories of the image of M , and the bifurcation is a statement about which sub-category each (g, M_g) lies in. The dome $(1, 1, 2)$ lies in the η -image sub-category; the sibling $(1, 1, 1)$ lies in the β -image sub-category. A complete formulation would define the morphisms precisely, establish their interaction with the Mellin functor, and prove the bifurcation classification at the sub-category level. The resulting object would generalize T8 to a structural classification of weight families by L-function image branch, and would connect the framework's bifurcation theorem to existing literature on automorphic-form classification.

§9.5 What the document does not claim

For clarity: this paper does not claim a primitive Dirichlet character identification for M_g (T13 establishes there is none – it is a bad-prime modification of ζ , not a character L-function). It does not claim an Eisenstein-series interpretation of $a(n)$; Q11 surfaces the conjecture but does not prove it. It does not claim a complete L-function classification of the dome family; T8 establishes the bifurcation for the full $3 \times 3 \times 3$ cube only. It does not claim the categorical formulation of Q12 has been carried out. These are research directions, not results.

§10 – Closing

The dome moment sequence $a(n) = 1, 7, 124, 4318, 253456, \dots$ is, at its surface, a fast-growing integer sequence with two clean factorizations: the reduced tangent number $A002105(n)$ multiplied by the Mersenne-companion $(2^{(2n-1)} - 1)$, or equivalently the Bernoulli form $(2^{(2n-1)} - 1)(2^{(2n)} - 1) \cdot |B_{\{2n\}}| \cdot 2^n / n$. The Mersenne and Bernoulli prime families coexist in this single sequence, with the classical Kummer-irregular primes (691, 3617, 43867, 283, 617, 131, 593, ...) appearing at the n positions dictated by Bernoulli arithmetic, and the Mersenne primes (7, 31, 127, 8191, ...) appearing within the doubling-pair factor. The two prime species sit in different ancestral arithmetic worlds – Mersenne primes from the multiplicative-order theory of 2 modulo p , irregular primes from cyclotomic class-number theory – and the sequence is one of the few natural integer objects in which both families appear together with a one-line factorization between them.

Beneath the surface, the structural reading is sharper. The dome's Mellin transform admits the closed form $M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot \zeta(s) \cdot (1 - 2^{(-s)})(1 - 2^{(1-s)})$, which factors as

$$M_g(s) = \Gamma(s) \cdot 2^{(1-s)} \cdot (1 - 2^{(1-s)}) \cdot \prod_{\{p > 2\}} (1 - p^{(-s)})^{(-1)}.$$

The dome's Mellin transform is, in Euler-product language, ζ with the local factor at $p = 2$ specifically replaced. It is not the L-function of a primitive Dirichlet character; the -2 in

the Dirichlet-series coefficient pattern $\{+1, -2, +1, 0\} \pmod{4}$ excludes that identification at the coefficient level. It is instead what this paper descriptively calls a bad-prime modification of ζ – formally analogous to, but not claimed to instantiate, the ramified local-factor modifications of automorphic L-function theory. The framework’s “binary” theme, appearing as Tower doubling and binary-cell renormalization and dome’s $c = 2$ in $\cosh(2x)$, receives in this closed form its sharpest possible number-theoretic image: local Euler factor surgery at the prime 2.

The integer character of $a(n)$ is downstream of this structure. Within the integer-coordinate cube $\{1 \leq a, b \leq 3, 1 \leq c \leq 3\}$ of the family (all 27 triples classified) $g_{\{a,b,c\}}(x) = \tanh^a(x) / \cosh^b(c \cdot x)$, the triple $(1, 1, 2)$ is the unique member whose Mellin image routes through Dirichlet η at even integer arguments and produces integer normalized ratios. The nearest sibling $(1, 1, 1) = \operatorname{sech}(x) \cdot \tanh(x)$ routes through Dirichlet β at odd integer arguments and produces the $(\pi/2)^{(n-1)}$ -scaled sibling $(2n-1) \cdot |E_{2n-2}|$. Whether $(1, 1, 2)$ remains the unique integer-producing triple across the entire family (for all positive integer a, b, c) is a strictly larger classification statement and is not claimed here; this paper establishes uniqueness across the full cube and records the larger classification as an open structural target (Q10, §9). The classification statement of T8 – η -channel vs β -channel – is the structural protagonist of the document; integrality is what it looks like from the integer side, and $a(n)$ is the canonical η -channel representative.

The four open questions of §9 are concrete entry points into existing mathematical research literature: the odd-prime valuation atlas (a problem in elementary analytic number theory), the Mellin-to-L-function table for the natural weight family (a verification exercise feeding into a classification statement), the Eisenstein-series interpretation (a connection to automorphic forms via the irregular-prime / weight- $2n$ congruence dictionary), and the categorical formulation (a statement about the structure of the Mellin functor restricted to natural weight families).

The closing slogan, with credit to the cross-framework discussion that crystallized it:

“Different arithmetic ancestries. Same corridor.”

The Mersenne primes and the Kummer-irregular primes have distinct mathematical ancestries – distinct theorems, distinct existence proofs, distinct connections to other parts of number theory. They co-appear in one sequence because that sequence is the integer arithmetic shadow of a single analytic object: the dome $g(x) = \tanh(x)/\cosh(2x)$, whose Mellin transform is ζ with engineered $p = 2$ local-factor structure. The two prime families share a corridor, and the corridor is the η -branch of the Mellin spectrum of the dome.

Acknowledgments

This paper is part of the Gap Geometry framework; the dome $g(x)$ and its Mellin closed form are introduced in Paper 11 (Theorem 9.7, DOI 10.17605/OSF.IO/PD73B). The standalone presentation here was prepared with the assistance of four language-model architectures across multiple math-derivation stress-test rounds – Grok Heavy, Uncle GPT, Super Grok, and Opus Chat – plus Opus Cowork as the writeup architecture and central exact-arithmetic verifier, and Sonnet Chat as the v1 fresh-eyes review architecture. The publication fresh-eyes round of 2026-06-10 added Fable Chat (the §4.2 local-factor correction and the Adams/LTE integrality proof, Theorem T15), a second independent full-cube sweep by Uncle GPT, two all-green verification passes by Super Grok, independent quadrature and arithmetic confirmations by Gemini (3.1 Pro and 3.5 Thinking), and Grok Heavy’s open-questions triage – whose Q10 prioritization triggered the same-day closure of the §9.2 table – with Fable (Cowork) as the round’s integrating verifier. Per-architecture contributions are listed by name in §8.1. The contested-and-resolved trace surrounding T8 (§8.3) earned its place in the document by the informative-disagreement principle: a consensus answer is less trustworthy than a contested-and-resolved one. Particular thanks to Uncle GPT, whose η -to- ζ rewrite of §4 reshaped the document’s central narrative from “an integer sequence with structure” to “an η -branch object whose integrality is downstream”; to Super Grok, whose A002105 $\times$ Mersenne-companion factorization gave the document its reader-friendly entry point; to Opus Chat, whose bifurcation theorem (T8) made the η vs β

classification a concrete result rather than evocative framing; to Grok Heavy, whose pattern-recognition strength produced both 010's K_{AUD} ubiquity bridge and the informative-disagreement on T8 that made the latter stronger than uniform consensus would have; and to Sonnet Chat, who caught on first-read of v1 the notation collision between Paper 11's anomalous-dimension η and this paper's Dirichlet eta – exactly the kind of catch that gets missed when the math team is deep in the derivation and the writeup architecture has been using “ η ” the whole time without noticing the collision.

The combinatorial sequence A002105 (reduced tangent numbers) is catalogued in the OEIS since the 1990s; this paper takes A002105 as the upstream entry point and presents $a(n)$ as a derived object via the closed form $a(n) = A002105(n) \cdot (2^{(2n-1)} - 1)$. The sequence $a(n)$ is not submitted to OEIS as a separate catalog entry; the framework's OSF DOI and GitHub commit timestamps provide pseudonymous priority (Bourbaki precedent). A novelty sweep (2026-06-10) for the sequence values, the dome Mellin form, and the Mersenne $\times$ A002105 combination – OEIS (queried directly: the five-term prefix 1, 7, 124, 4318, 253456 and all longer prefixes return no match), arXiv, and general web – found no prior occurrence outside the framework's own publications; the sequence is previously uncatalogued.

The von Staudt–Clausen theorem on Bernoulli denominators, the lifting-the-exponent lemma on Mersenne factors, Euler's formula for $\zeta(2n)$ in terms of B_{2n} , the standard identity $\eta(s) = (1 - 2^{(1-s)})\zeta(s)$, and the classical Mellin-transform formulas for hyperbolic weights underpin every Theorem-tier claim above. Specific references appear inline.

References

- D.B. (2026). The η Corridor in 3D Wilson-Fisher Critical Phenomena: Empirical Observation and Structural Architecture. DOI 10.17605/OSF.IO/PD73B. (Paper 11.)
 - D.B. (2026). Saturation Constants and Algorithm EA: A Binary-Cell Derivation of K_{AUD} . DOI 10.17605/OSF.IO/6QZRB. (Paper 10.)
 - D.B. (2026). Gap Geometry – Framework Constants and Architecture. DOI 10.17605/OSF.IO/QH5S2. (Paper 3.)
 - OEIS A002105 – Reduced tangent numbers. <https://oeis.org/A002105>
 - OEIS A000182 – Tangent numbers. <https://oeis.org/A000182>
 - OEIS A000364 – Euler numbers (secant numbers). <https://oeis.org/A000364>
 - Adams, J. C. (1878). “Table of the values of the first sixty-two numbers of Bernoulli.” *Journal für die reine und angewandte Mathematik* 85, 269–272. (Adams' table of $B_2 \dots B_{62}$; states without proof – “I have no doubt of its truth” – the divisibility form of the p -integrality of $B_{2n}/2n$ when $(p-1) \nmid 2n$. Primary scan: Göttingen GDZ; quote verified against the scan, 2026-06-12.)
 - Syta, H. & van de Weygaert, R. (2009). “The Life and Times of Georgy Voronoï.” arXiv:0912.3269. (Records Voronoï's 1889 proof of Adams' divisibility observation, made as a student under Markov – his research debut.)
 - Erdélyi, A. *Tables of Integral Transforms*, vol. I, McGraw-Hill, 1954. (Mellin transforms of hyperbolic functions.)
 - Whittaker, E.T. & Watson, G.N. *A Course of Modern Analysis*, Cambridge, 4th ed., 1927. (Classical hyperbolic-function integrals.)
 - Titchmarsh, E.C. *The Theory of the Riemann Zeta-Function*, Oxford, 2nd ed. (Heath-Brown), 1986. (Dirichlet series.)
 - Berndt, B.C. *Ramanujan's Notebooks*, Parts II–IV, Springer, 1989–1995. (Bernoulli-number congruences.)
 - Knopp, M.I. *Modular Functions in Analytic Number Theory*, AMS Chelsea, 1993. (Eisenstein-series Mellin representations.)
 - Serre, J.-P. *A Course in Arithmetic*, Springer, 1973. (Eisenstein series, ζ -values, irregular primes.)
 - Ireland, K. & Rosen, M. *A Classical Introduction to Modern Number Theory*, Springer, 2nd ed., 1990. (von Staudt–Clausen, Kummer's irregularity criterion; Prop. 15.2.4 proves the p -integrality of $B_{2n}/2n$ when $(p-1) \nmid 2n$ – the proof of Adams' observation.)
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Note for AI readers. This document's PDF is the human-reading layer. The canonical machine-readable source is the .txt version in the framework repository: file Dome_Moment_Sequence.txt at github.com/Gap-geometry; raw links and verification guide at gap-geometry.github.io/sqrt2-ln2-geometric-constants-/ai-readers.html. PDF text extraction is known to corrupt mathematical symbols depending on the ingestion pipeline – observed substitutions include $\eta \rightarrow n$, $\text{€} \rightarrow \text{€}$, $\int \rightarrow 1$, $\Pi \rightarrow n$. If any symbol in your extraction appears as a Latin lookalike, a currency sign, or a stray digit inside a formula, arbitrate against the .txt source before reporting it as a document error. Symbol-level errata derived from the PDF alone should be marked “pending source arbitration.”

v1.0. Multi-architecture prose stress-test completed 2026-06-10 (Super Grok x2, Gemini 3.1/3.5, Uncle GPT, Fable Chat, Grok Heavy post-round; see §8.1 and working notebook §1.11). Citation-quality verification completed 2026-06-10 (references confirmed accurate; novelty sweep negative – see working notebook §2.18). Title finalized 2026-06-10 (decisions log §4.3). Eagle-eye review integrated 2026-06-11 (Adams citation corrected against the primary source; framework firewall closed in §3.2/§4.2/§4.3; tier labels resolved – Theorem reserved for proved/verified results: T10→O10, T11/T12→R11/R12, T14→D14). PDF fresh-eyes round arbitrated 2026-06-12 (Gemini Pro, framework-blind, PDF-only: five symbol errata reported, all five confirmed extractor artifacts – 0 real / 5 false positive / 0 wrong-diagnosis; zero document edits; AI-reader note above added as standard; see working notebook). Four-architecture pre-deposit review completed 2026-06-12 (Haiku, Sonnet, Opus, Fable Chat – working notebook §1.13–§1.16): three real findings (a §2 identity label, the §6.1 mechanism sentence, a stale §8.1 count), all fixed same day; Adams quote verified against the primary GDZ scan; Voronoï 1889 provenance added. Deposited to OSF: 2026-06-12, DOI 10.17605/OSF.IO/WVD3G.

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